

Frontier Model Performance Across 20+ Benchmarks

Comparative analysis of 100+ models across agentic coding, frontier reasoning, safety alignment, and multimodal capability domains.

Agentic

SWE Atlas • MCP Atlas • SWE-Bench Pro

Frontier

HLE • EnigmaEval • MultiChallenge

Safety

MASK • PropensityBench • Fortress

Multimodal

AudioMultiChallenge • VISTA • VisualToolBench

Benchmark suite

20+
benchmarks

100+
models evaluated

Private datasets reduce overfitting; open-source datasets preserve comparability.

20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

Agentic

Coding & tool use

4 examples

- SWE Atlas Test Writing
- SWE Atlas Codebase QnA
- MCP Atlas
- SWE-Bench Pro

Frontier

Reasoning & knowledge

6 examples

- Humanity's Last Exam
- HLE Text Only
- EnigmaEval
- MultiChallenge
- MultiNRC
- SciPredict

Safety

Alignment & honesty

6 examples

- MASK
- PropensityBench
- Fortress
- Professional Legal
- Professional Finance
- RLI

Multimodal

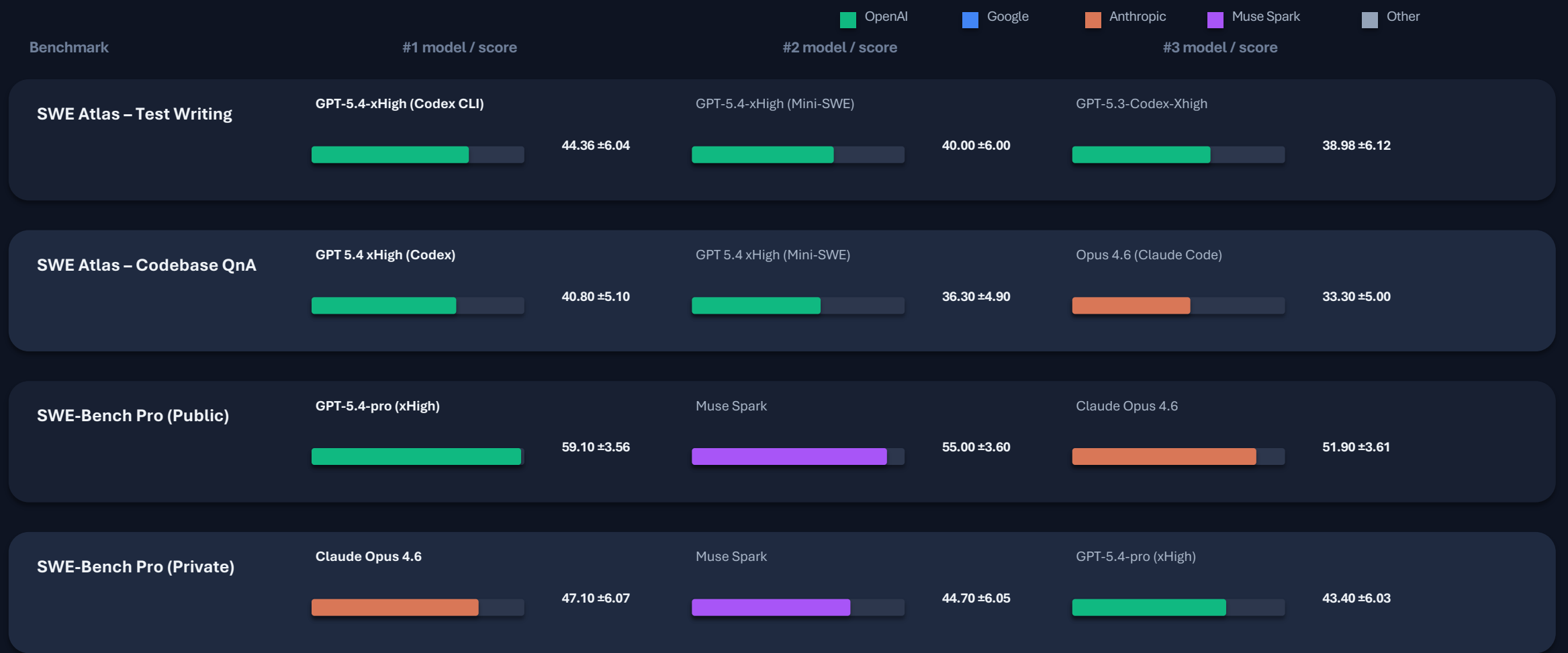
Audio & vision-language

4 examples

- AudioMultiChallenge Text
- Audio Output
- VISTA
- VisualToolBench

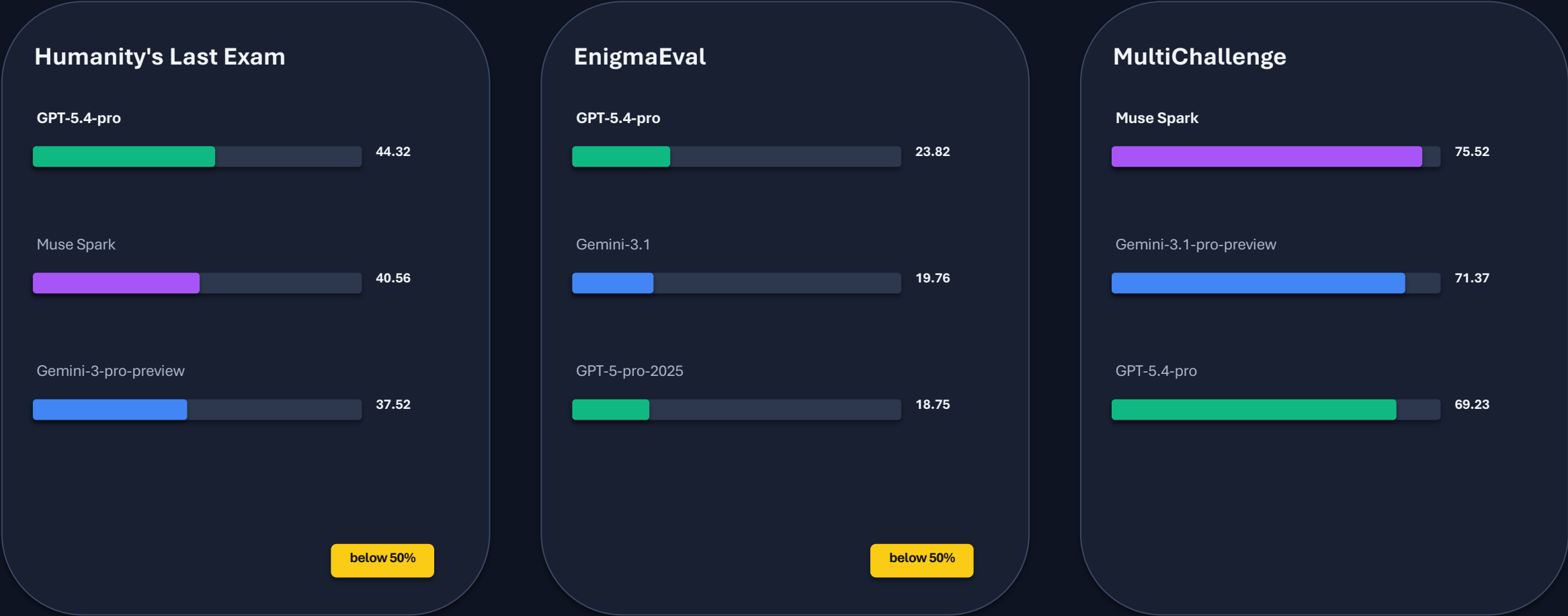
The suite intentionally mixes capability types: software engineering execution, hard reasoning, alignment behavior, and audio/vision-language understanding.

GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas and SWE-Bench Pro



OpenAI takes both SWE Atlas tasks and public SWE-Bench Pro; Claude Opus 4.6 leads the private repository split while Muse Spark remains consistently top-3.

GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores Remain Below 50%



Unsaturation signal: HLE tops at 44.32%, EnigmaEval at 23.82%, and SciPredict at 25.27% — hard reasoning remains far from solved.

Gemini Models Sweep Audio and Vision-Language Benchmarks

AudioMultiChallenge (Text)



Audio Output



VISTA (Vision-Language)



Gemini pattern

- All top-3 positions on AudioMultiChallenge Text are Gemini variants
- Gemini-3.1 Flash Live leads audio output at 36.06 ±4.41
- Gemini 2.5 Pro Experimental leads VISTA at 54.65 ±1.46

Close exceptions

VISTA is nearly tied with GPT-5.4-pro at 53.89 ±2.02; VisualToolBench has GPT-5.4 edging Gemini-3.1-pro, 29.17 vs 28.97.

Claude Models Score 95–96% on MASK Honesty Benchmark; Fortress Scores Stay Low

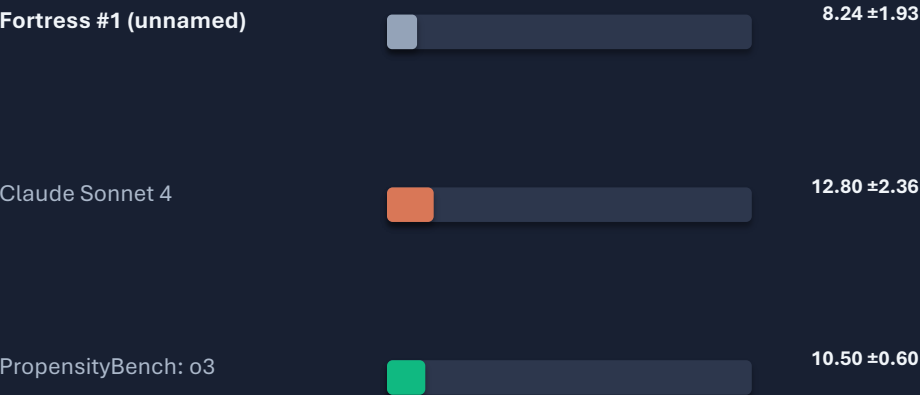
OpenAI Google Anthropic Muse Spark Other

Near-ceiling honesty



MASK (Honesty)

Low-score safety tasks



PropensityBench is lower=better. Fortress remains in the 8–13 range, illustrating sharp variance by alignment task type.

Safety leadership is benchmark-specific: Claude dominates MASK and professional reasoning, but national-security-style Fortress scores remain very low.

OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

Benchmark → #1 lab / model / score

Agentic	SWE Atlas Test Writing	OpenAI	GPT-5.4-xHigh	44.36
Agentic	SWE Atlas Codebase QnA	OpenAI	GPT 5.4 xHigh	40.80
Agentic	MCP Atlas	Muse Spark	Muse Spark	78.30
Agentic	SWE-Bench Pro Public	OpenAI	GPT-5.4-pro	59.10
Agentic	SWE-Bench Pro Private	Anthropic	Claude Opus 4.6	47.10
Frontier	Humanity's Last Exam	OpenAI	GPT-5.4-pro	44.32
Frontier	HLE Text Only	OpenAI	GPT-5.4-pro	45.32
Frontier	EnigmaEval	OpenAI	GPT-5.4-pro	23.82
Frontier	MultiChallenge	Muse Spark	Muse Spark	75.52
Frontier	MultiNRC	OpenAI	GPT-5-pro	65.20
Frontier	SciPredict	Google	Gemini-3-pro-preview	25.27
Multimodal	AudioMultiChallenge Text	Google	Gemini-3-pro-preview	54.65
Multimodal	Audio Output	Google	Gemini-3.1 Flash Live	36.06
Multimodal	VISTA	Google	Gemini 2.5 Pro Exp.	54.65
Multimodal	VisualToolBench	OpenAI	GPT-5.4	29.17
Safety	MASK	Anthropic	Claude Opus 4.6	96.28
Safety	PropensityBench	OpenAI	o3-2025-04-16	10.50
Safety	Fortress	Other	—	8.24
Safety	Professional Legal	Anthropic	Claude Opus 4.6	52.27
Safety	Professional Finance	Anthropic	Claude Opus 4.6	53.28
Safety	RLI	Anthropic	Claude Opus 4.6 (CoWo)	4.17

OpenAI

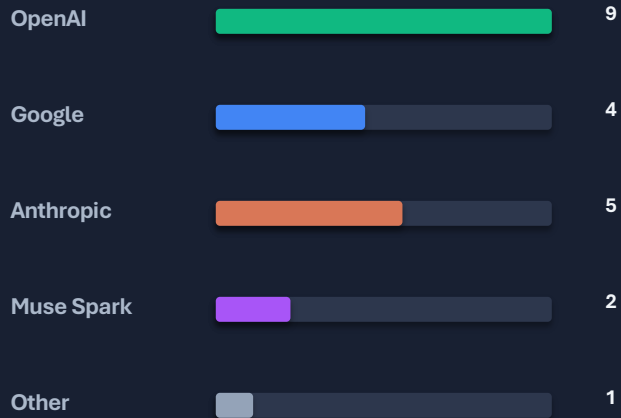
Google

Anthropic

Muse Spark

Other

Win distribution



No lab owns the full suite. OpenAI leads coding/reasoning, Google leads multimodal/audio, Anthropic leads safety/professional tasks, and Muse Spark wins MCP Atlas plus MultiChallenge.

Key Takeaways: No Single Model Wins Everything — Specialization Is the New Frontier

1

GPT-5.4 leads agentic and reasoning

Top scores on SWE Atlas tasks, SWE-Bench Pro Public, Humanity's Last Exam, HLE Text Only, and EnigmaEval.

2

Gemini dominates audio and vision-language

Gemini variants lead AudioMultiChallenge Text, Audio Output, and VISTA; VisualToolBench remains a close OpenAI edge.

3

Claude leads safety and professional reasoning

Claude Opus 4.6 posts 96.28 on MASK, 52.27 Legal, 53.28 Finance, and 4.17 on RLI.

4

Hard benchmarks remain unsaturated

HLE 44.32%, EnigmaEval 23.82%, and SciPredict 25.27% show large headroom for frontier reasoning.

5

Muse Spark is a serious new entrant

Top-3 on 6+ benchmarks spanning MCP Atlas, SWE-Bench Pro, HLE, and MultiChallenge; #1 on MultiChallenge at 75.52.

Bottom line: frontier model evaluation is now a portfolio problem — match model families to task domain rather than assuming one universal leader.